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clc; clear; close all;

%% (i) Signal & Sampling
f = 6; % Signal frequency (R = 6)
Fs = 8*f; % Sampling frequency
t = 0:1/Fs:1;
x = sin(2*pi*f*t); % Sampled discrete-time signal

% (ii)  $\mu$ -law Companding Quantization
N = 8; % Number of bits
L = 2^N;
mu = 100;
V = max(abs(x));

x_norm = x/V;
x_comp = compand(x_norm,mu,1,'mu/compressor');

step = (max(x_comp)-min(x_comp))/(L-1);
xq = round(x_comp/step)*step;

% (iii) Encoding
encoder = dsp.UniformEncoder('PeakValue',1,'NumBits',N);
decoder = dsp.UniformDecoder('PeakValue',1,'NumBits',N);

encoded = encoder(xq);
decoded = decoder(encoded);

% (iv) Expansion & Reconstruction
x_exp = compand(decoded,mu,1,'mu/expander');
x_exp = x_exp * V;

t_rec = 0:0.001:1;

zoh = interp1(t,x_exp,t_rec,'previous'); % Zero Order Hold
foh = interp1(t,x_exp,t_rec,'linear'); % First Order Hold

% Plots
figure;
subplot(3,1,1)
plot(t,x,'k','LineWidth',1.5)
title('Sampled Signal x[n]')
xlabel('Time'); ylabel('Amplitude')

subplot(3,1,2)
plot(t,x_exp,'r','LineWidth',1.5)
title('Quantized & Decoded Signal')
xlabel('Time'); ylabel('Amplitude')

subplot(3,1,3)
plot(t_rec,foh,'b',t_rec,zoh,'g--','LineWidth',1.2)
legend('FOH','ZOH')
title('Reconstructed Signals')
xlabel('Time'); ylabel('Amplitude')
grid on

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